AUTOMATIC DETECTION OF PARKINSON'S DISEASE FROM VERTICAL GROUND REACTION FORCE SIGNALS USING TSFRESH AND CLASSICAL MACHINE LEARNING

Saranya Bhattacharjee (aj81owid)

Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany

ABSTRACT

This report outlines a TSFresh-based feature engineering pipeline coupled with classical machine learning algorithms for Parkinson's disease (PD) versus healthy control (HC) classification using bilateral vertical ground reaction force (vGRF) signals. Preliminary data analysis and exploratory visualization were conducted to characterize gait dynamics. Window-level time-series features were extracted and aggregated at the subject level using distributional statistics (mean, standard deviation, skewness, and kurtosis) to obtain robust subject representations. Three feature configurations–left-foot, right-foot, and bilateral combined–were evaluated under nested stratified five-fold cross-validation with random forest, SVM-linear, and SVM-RBF. The combined bilateral SVM-RBF achieved the best AUC of 0.85 ± 0.04, demonstrating that bilateral fusion and nonlinear classifiers are complementary. Left-foot models showed high sensitivity but poor specificity, consistent with asymmetric compensatory loading in PD. The code is available at https://github.com/saranyab21/Project-Time-Series-WS-25-26.

***Index Terms*—** Parkinson's disease, vGRF, TSFresh, nested cross-validation, PCA, random forest, SVM

1. INTRODUCTION

Parkinson's disease (PD) is the second most widespread neurodegenerative disorder worldwide, affecting the motor system predominantly, leading to symptoms such as tremors, muscle rigidity, and poor balance and coordination [1]. As the disease progresses, gait impairments directly increase fall risk and degrade quality of life.

Recent research has increasingly leveraged biomechanical data, particularly objective sensor-based biomarkers derived from vertical ground reaction force (vGRF) signals, to analyze gait behavior in individuals with Parkinson's disease. These signals encode high-resolution information about loading patterns, stride symmetry, and propulsion, all of which are frequently disrupted in PD [2].

MSc Artificial Intelligence, Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg. Supervisor: Dr.-Ing. Tomás Arias-Vergara.

1.1. Related Work

Existing approaches to automated PD detection from gait signals vary considerably in sensor modality, feature design, and validation strategies. Trabassi et al. [3] utilised a manual feature selection pipeline through statistical filtering and sequential backward elimination and achieved a classification accuracy exceeding 80%; however, the single-sensor setup failed to capture bilateral gait asymmetries. In [4], the authors extracted temporal gait parameters per leg independently using gyroscope peak detection, meaning inter-limb coordination was structurally absent from the feature space, and bilateral gait asymmetry went unrepresented. In contrast, [5] employed the same PhysioNet vGRF dataset used in this work, with Random Forest and SMOTE oversampling. However, optimistic bias is introduced when SMOTE is applied before cross-validation splitting, and hyperparameter tuning and generalization estimation are confused when nested cross-validation is not used.

1.2. Contributions

This report presents: **(i)** a complete pre-processing and descriptive analysis of the PhysioNet gait database, including sex-stratified UPDRS statistics and illustrative signal plots; **(ii)** a sliding-window TSFresh extraction pipeline with subject-level aggregation for left-foot, right-foot, and combined configurations; **(iii)** PCA-based visualisation of feature class structure; and **(iv)** a nested 5-fold cross-validation comparison of RF, SVM-Linear, and SVM-RBF, with all preprocessing conducted within each fold to prevent data leakage.

2. MATERIALS & METHODS

2.1. Dataset

The PhysioNet *Gait in Parkinson's Disease* database [6] contains 166 subjects: 93 PD (58M/35F; age 66.3 ± 9 yr; UPDRS 31 ± 12) and 73 age-matched HC (40M/33F; age 66.3 ± 9 yr). vGRF was sampled at 100 Hz from 8 load sensors per foot (16 channels total) during usual-walk trials only; dual-task trials were excluded as cognitive load inflates stride variability independently of motor impairment. All trials per subject were

concatenated before processing. A Mann–Whitney U test on UPDRS by sex (PD cohort only) yielded $p = 0.049$, Cliff's $\delta = 0.25$ (small effect), indicating mildly higher and more variable impairment in male PD subjects (as shown in Fig. 1). However, this has no meaningful impact on the gait classification task.

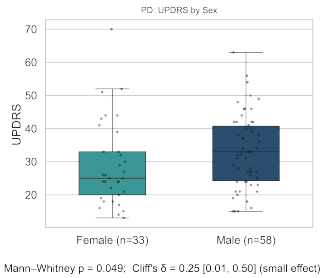

Fig. 1: UPDRS score distribution by sex within the PD cohort.

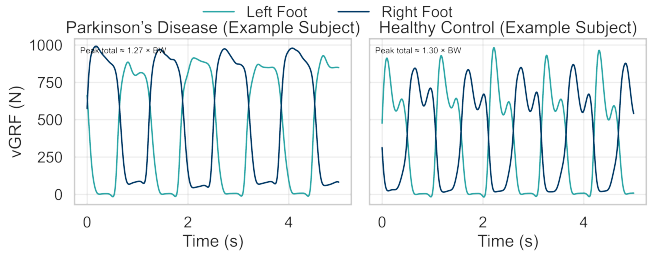

Fig. 2: Five-second filtered vGRF excerpt (left and right foot,usual-walk trial,gait-initiation period excluded) for a PD subject (SiPt29, UPDRS = 70, 0.86 m/s) and an HC subject (JuCo17, 1.54 m/s). PD shows lower peak forces, reduced inter-limb symmetry, and irregular step timing.

2.1.1. Pre-processing

Each 16-channel file was reduced to two signals by summing the 8 sensors per foot (Eq. 1), giving the total vGRF per foot rather than individual pressure readings at each sensor location. A zero-phase second-order Butterworth low-pass filter at 6 Hz was applied via `scipy.signal.filtfilt` to remove high-frequency noise while preserving all physiologically relevant gait harmonics. Where body mass was unavailable, body weight (BW) was estimated from the median total vGRF during stance phases, defined as time points where the bilateral total exceeded 100 N.

$$\mathrm{vGRF}_{\mathrm{foot}}(t) = \sum_{i=1}^{8} s_i(t), \quad \widetilde{\mathrm{vGRF}}(t) = \frac{\mathrm{vGRF}(t)}{m \cdot g} \tag{1}$$

Figures 2 and 3 illustrate the filtered signals and BW-normalised mean stride profiles for one PD and one HC example subject, respectively.

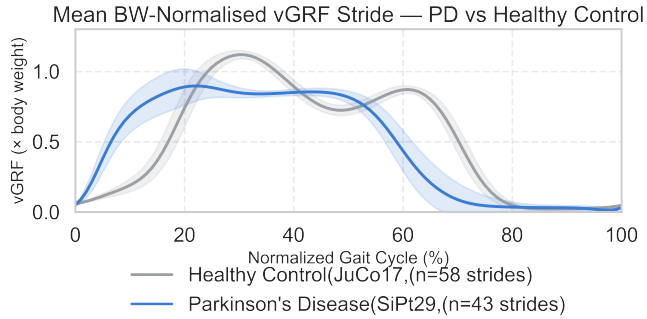

Fig. 3: Mean BW-normalised vGRF stride (± 1 SD shading) over the normalised gait cycle (0–100%), right foot. HC retains the canonical two-peak pattern; PD shows flattened, merged peaks and wider variability bands, indicating impaired propulsion and reduced stance-control consistency.

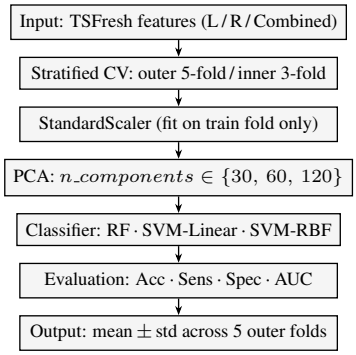

Fig. 4: End-to-end classification pipeline. All preprocessing steps (StandardScaler → PCA → Classifier) are encapsulated inside each outer fold to prevent data leakage. Inner 3-fold GridSearchCV tunes PCA components $\in \{30, 60, 120\}$ and classifier hyperparameters jointly, scored by ROC AUC.

2.2. Proposed Approach

Figure 4 summarises the pipeline.

2.2.1. TSFresh Feature Extraction & Aggregation

Each filtered signal was segmented with a 300 ms sliding window (30 samples at 100 Hz) and 150 ms step (50% overlap). TSFresh `ComprehensiveFCParameters` was applied per window; columns containing only NaN or infinite values were removed. Window-level features were then aggregated per subject across all windows using mean, standard deviation, skewness, and kurtosis, producing one feature vector per subject per foot. Three feature matrices were formed: **Left** (L_ prefix), **Right** (R_ prefix), and **Combined** (L + R concatenated).

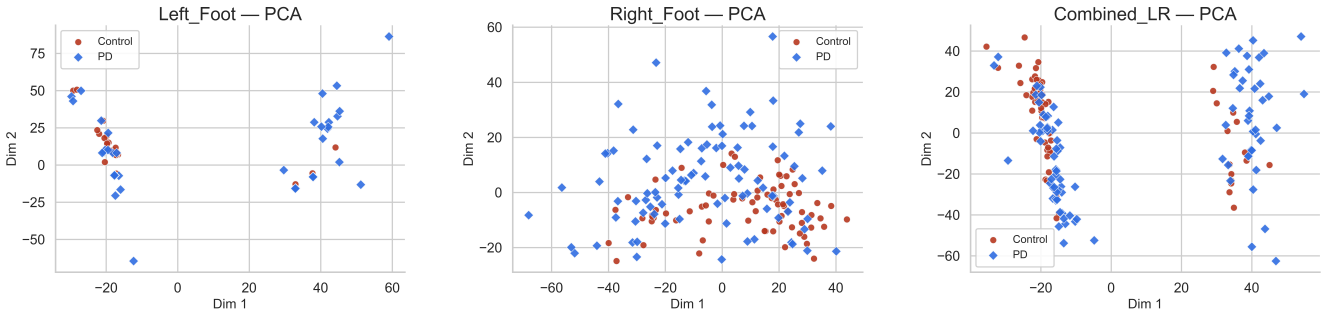

Fig. 5: PCA scatter plots (PC1 vs. PC2) for left-foot (left), right-foot (centre), and combined (right) feature configurations. Blue diamonds: PD; red circles: HC.

2.2.2. *Classifiers*

Three baselines were evaluated: **RF** (ensemble size $\in \{200, 400, 800\}$, max depth $\in \{\text{None}, 10, 20, 40\}$, min samples split $\in \{2, 5, 10\}$); **SVM-Linear** ($C \in \{0.01, 0.1, 1, 10, 100\}$); and **SVM-RBF** ($C \in \{0.01, \ldots, 100\}$, $\gamma \in \{\text{scale}, 10^{-4}, 10^{-3}, 10^{-2}\}$).

2.2.3. *Evaluation & Training Strategy*

Nested stratified cross-validation was used throughout: an *outer* 5-fold loop estimates generalisation; within each outer training partition an *inner* 3-fold `GridSearchCV` optimises all pipeline hyperparameters (PCA dimensionality and classifier parameters) using ROC AUC. With only 166 subjects, using standard k-fold and tuning hyperparameters on the same data would leak information into the test fold and give overly optimistic results, so nested CV was used to keep the test fold completely held out. Reported metrics are accuracy, sensitivity (recall for PD), specificity (recall for HC), and AUC, as mean $\pm$ std across five outer folds.

3. EXPERIMENTS & RESULTS

Three feature configurations (Left, Right, Combined) were each evaluated against three classifiers under nested CV. Results are reported as mean $\pm$ std across five outer folds.

3.1. Experiment 1 - PCA Feature Visualisation

Figure 5 shows 2D PCA projections (PC1 vs. PC2) of the z-score normalised feature matrices for all three configurations. In the left-foot plot, PD subjects appear more spread out than controls, which likely reflects higher stride variability on the more affected side. The right foot projection shows some separation along PC1, but the groups still overlap considerably. The combined plot is the most interesting: the right-side cluster is mostly PD subjects, which at least suggests that combining both feet picks up something the individual feet do not. Overall though, the groups are not cleanly separable in 2D, which is expected since PCA does not optimise for class boundaries, so discriminative structure may exist in later components that the classifier can still exploit.

3.2. Experiment 2 - Baseline Classification Performance

Table 1 summarises nested CV metrics for all nine model–feature combinations. AUC improves monotonically Left $<$ Right $<$ Combined across all classifiers; the bilateral AUC gain over left-foot alone is +0.17–0.20 (shown in Fig. 6. Nonlinear models (RF, SVM-RBF) outperform SVM-Linear in every configuration. The left-foot configuration exhibits a pronounced imbalance: SVM-RBF achieves sensitivity 0.89 ± 0.11 but specificity only 0.45 ± 0.24, indicating that the left-foot signal alone is not sufficient for reliable discrimination. The combined SVM-RBF achieves the best result: Acc 0.78 ± 0.05, Sens 0.78 ± 0.09, Spec 0.72 ± 0.11, AUC $\mathbf{0.85 \pm 0.04}$.

Table 1: Nested cross-validated performance (mean $\pm$ std, 5 outer folds). Sens.: recall for PD. Spec.: recall for HC. Best overall AUC in **bold**.

Features	Model	Acc.	Sens.	Spec.	AUC
Left	RF	0.64 ± 0.07	0.75 ± 0.11	0.49 ± 0.11	0.68 ± 0.05
	SVM-Lin	0.64 ± 0.08	0.83 ± 0.07	0.40 ± 0.09	0.62 ± 0.11
	SVM-RBF	0.67 ± 0.07	0.89 ± 0.11	0.45 ± 0.24	0.65 ± 0.05
Right	RF	0.69 ± 0.04	0.83 ± 0.07	0.52 ± 0.13	0.76 ± 0.08
	SVM-Lin	0.67 ± 0.05	0.78 ± 0.10	0.54 ± 0.11	0.75 ± 0.07
	SVM-RBF	0.72 ± 0.05	0.76 ± 0.05	0.65 ± 0.13	0.78 ± 0.05
Combined	RF	0.73 ± 0.03	0.82 ± 0.08	0.62 ± 0.06	0.85 ± 0.07
	SVM-Lin	0.75 ± 0.04	0.75 ± 0.07	0.75 ± 0.11	0.82 ± 0.06
	SVM-RBF	0.78 ± 0.05	0.78 ± 0.09	0.72 ± 0.11	$\mathbf{0.85 \pm 0.04}$

3.3. Experiment 3 - Confusion Matrix Analysis

Across configurations the confusion matrices (in Fig. 7) confirm the quantitative trends in Table 1. For the left-foot, all three classifiers exhibit the same pattern: high PD recall but consistently poor HC specificity. Since this limitation persists

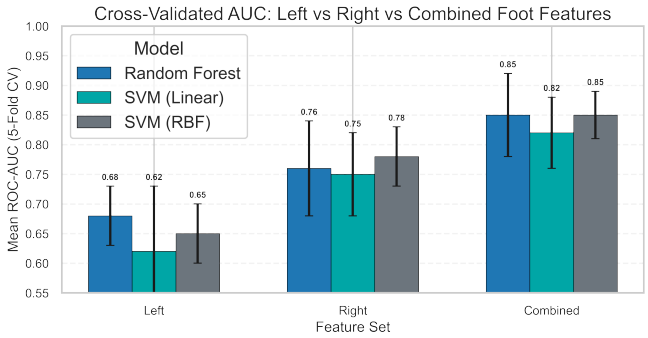

Fig. 6: Mean ROC-AUC (± 1 SD error bars) across all feature configurations and classifiers. AUC increases Left < Right < Combined for all models; RF and SVM-RBF dominate; SVM-Linear underperforms consistently, confirming a nonlinear decision boundary.

regardless of model complexity, it is reasonable to attribute it to the left-foot signal rather than to model choice. The right foot tells a different story - RF achieves near-perfect separation across both classes, indicating that the right-foot TSFresh features carry strong discriminative information. SVM-RBF performs well but shows some residual errors, while SVM-Linear struggles noticeably, which points to a nonlinear decision boundary in this feature space. The combined configuration produces the most consistent results: RF makes almost no errors, SVM-RBF remains balanced across both PD and HC, and SVM-Linear improves relative to the single-foot cases but still underfits the expanded bilateral feature space. Overall, using both feet together clearly helps with the HC misclassification problem, and nonlinear models are needed to make the most of the combined features.

4. DISCUSSION

The key takeaways from the experiments are as follows:

- Even though the PCA plots show overlap between PD and HC, the classifiers still perform well because they use far more components than the 2 shown in the scatter plots, with the optimal number selected per fold during training.
- The right foot being more discriminative than the left makes sense — PD patients often favour one side, so the other foot ends up carrying more weight and shows clearer gait differences.
- SVM-Linear failing across all three feature sets suggests the features do not separate linearly, which is why RF and SVM-RBF work better here.
- Despite fully automated feature extraction, the pipeline reaches comparable performance to [5] on the same dataset, while avoiding SMOTE leakage and absent nested validation that inflate their reported accuracy.
- The main limitations are the small dataset ($n = 166$) and window aggregation discarding temporal ordering - summary statistics cannot capture how gait patterns evolve across a full walking trial.

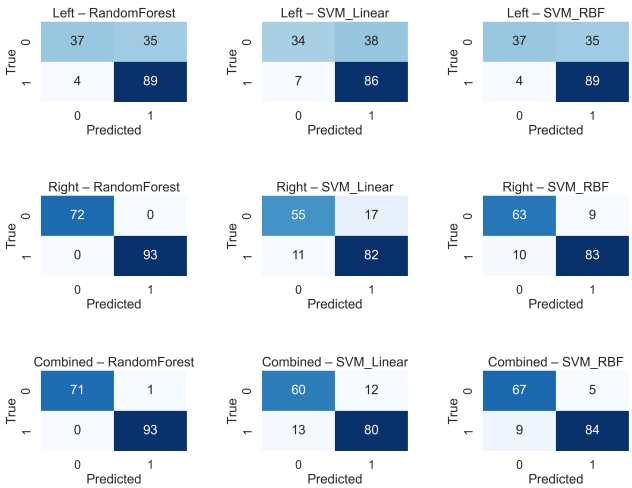

Fig. 7: Confusion matrices for all feature configurations (rows: Left, Right, Combined) and classifiers (columns: RF, SVM-Linear, SVM-RBF). Fitted on the full dataset via inner GridSearchCV.

5. CONCLUSION

This work presented a vGRF-based PD detection framework with a sliding-window TSFresh extraction followed by a nested stratified cross-validation with a joint PCA-classifier hyperparameter optimisation. The overall findings are: **(i)** bilateral feature fusion raises AUC by +0.17–0.20 over unilateral configurations, with the best result of 0.85 ± 0.04 (Combined, SVM-RBF); **(ii)** left-foot models are diagnostically asymmetric, displaying high PD sensitivity and poor HC specificity; **(iii)** nonlinear classifiers (RF, SVM-RBF) are necessary, as the TSFresh feature space lacks a competitive linear boundary; and **(iv)** nested CV is non-negotiable given $n = 166$ and a high-dimensional TSFresh feature space, where standard k-fold with internal tuning would produce optimistically biased estimates.

Future work will explore Mamba state-space models [7] on the raw temporal sequences to recover gait dynamics lost through aggregation, and validation on larger multi-site datasets. Moreover, implementing representation learning through self-supervised or deep embedding techniques to learn richer gait signatures can be explored.

6. REFERENCES

[1] Jinqiao Zhu, Yusha Cui, Junjiao Zhang, Rui Yan, Dongning Su, Dong Zhao, Anxin Wang, and Tao Feng, "Temporal trends in the prevalence of parkinson's disease from 1980 to 2023: a systematic review and meta-analysis," *The lancet Healthy longevity*, vol. 5, no. 7, pp. e464–e479, 2024.

[2] Qinghui Wang, Wei Zeng, and Xiangkun Dai, "Gait classification for early detection and severity rating of parkinson's disease based on hybrid signal processing and machine learning methods," *Cognitive Neurodynamics*, vol. 18, no. 1, pp. 109–132, 2024.

[3] Dante Trabassi, Mariano Serrao, Tiwana Varrecchia, Alberto Ranavolo, Gianluca Coppola, Roberto De Icco, Cristina Tassorelli, and Stefano Filippo Castiglia, "Machine learning approach to support the detection of parkinson's disease in imu-based gait analysis," *Sensors*, vol. 22, no. 10, pp. 3700, 2022.

[4] Vassilis Tsakanikas, Adamantios Ntanis, George Rigas, Christos Androutsos, Dimitrios Boucharas, Nikolaos Tachos, Vasileios Skaramagkas, Chariklia Chatzaki, Zinovia Kefalopoulou, Manolis Tsiknakis, et al., "Evaluating gait impairment in parkinson's disease from instrumented insole and imu sensor data," *Sensors*, vol. 23, no. 8, pp. 3902, 2023.

[5] Navita, Pooja Mittal, Yogesh Kumar Sharma, Anjani Kumar Rai, Sarita Simaiya, Umesh Kumar Lilhore, and Vimal Kumar, "Gait-based parkinson's disease diagnosis and severity classification using force sensors and machine learning," *Scientific Reports*, vol. 15, no. 1, pp. 328, 2025.

[6] Ary L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," `https://physionet.org/content/gaitpdb/1.0.0/`, 2000.

[7] Albert Gu and Tri Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv preprint arXiv:2312.00752*, 2023.